Assignment 1 - Group 5

Financial Engineering: Risk Management

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Introduction

This assignment focuses on evaluating and hedging an interest rate derivative portfolio as of February 15, 2008. The portfolio comprises a €650M ATM long payer swaption (1m10y expiry, 5y tenor) and a €550M vanilla 10-year receiver IRS. We compute the initial Mark-to-Market (MtM) and risk sensitivities (DV01) using the bootstrapped curve, and subsequently implement delta-hedging strategies against parallel and non-parallel curve shifts.

To execute these tasks, we utilized and corrected two main Python modules:

- `date_functions.py`: handles day-count conventions and payment schedules. We noted that the provided code only implements the Following convention; however, this does not impact our results because all payments fall in the middle of the month and never roll into the next month. We also kept the fixed leg as Adjusted for simplicity; although this is not the standard financial choice, it was not explicitly requested otherwise.
- `ex1_utilities.py`: implements the core financial models, including Black's formula for swaption pricing, exact swap valuation (MtM, par rates, BPV), and analytical risk approximations.

Task 1: Mark-to-Market the Portfolio

In this task, we compute the initial Mark-to-Market (MtM) of the portfolio. We assume the €550M 10-year receiver IRS is negotiated at par today. Therefore, its initial MtM is exactly zero. As a result, the total portfolio value depends exclusively on the payer swaption premium.

Description	**Result**
Forward Swap Rate (ATM Strike)	5.300%
Swaption Price (per unit notional)	€ 0.115955
10y Receiver IRS MtM (€550M)	€ 0.000000
Total Portfolio MtM	**€ 75,370,456.41**

Task 2: Portfolio DV01 (Parallel)

In this task, we evaluate the exact numerical DV01 by shifting the rates of all market instruments upwards by 1 basis point (+1 bp) and re-bootstrapping the zero curve. The total portfolio DV01 is **€ -367,999.97**.

This overall sensitivity is negative because the two instruments react in opposite directions. When market rates rise, the payer swaption gains value, while the receiver IRS loses value. The negative

impact of the IRS is larger than the positive impact of the swaption, driving the total portfolio into a loss.

Task 3: Analytical DV01 Approximation

In this task, we estimate the portfolio DV01 using an analytical approximation (Macaulay duration for the IRS and Black's delta for the swaption) and compare it with the exact numerical DV01 from Task 2.

Description	**Result**
Portfolio Proxy DV01	€ -294,501.17
Portfolio Numeric DV01	€ -367,999.97
Difference	**€ 73,498.80**

Commentary:
The analytical proxy underestimates the total portfolio loss. This discrepancy is mainly caused by the swaption delta approximation. Black's delta assumes that the discount factors (and therefore the Basis Point Value, BPV) remain constant when the forward swap rate increases.

In reality, when market rates go up, the discount factors drop. This reduces the BPV, which has a negative impact on the actual swaption price. Because the proxy ignores this BPV reduction, it overestimates the swaption's profit, making the total portfolio DV01 look more positive than it actually is.

Task 4: Delta-Hedge with a 10y IRS

In this task, we calculate the required notional of a 10-year IRS to delta-hedge the portfolio against parallel curve shifts. The market trades IRS in multiples of €1 million. We compute both the exact numerical hedge and the analytical hedge (proxy).

Note on Conventions:
For mathematical consistency, all hedging swaps in this report are modeled as receiver swaps. A negative notional represents a short position in a receiver swap (equivalent to a **payer**), while a positive notional represents a long position (**receiver**).

Description	**Result**
Numerical Hedge Notional (10y)	€ -459,000,000
Numerical Residual DV01	€ 31.00
Analytical Hedge Notional (10y)	€ -356,000,000

Commentary:
Since the portfolio DV01 is negative, we must take a short position in the 10-year receiver IRS (indicated by the negative notional) to neutralize the risk. The numerical hedge successfully reduces the residual portfolio DV01 to almost zero (only €31 due to the €1M rounding).

The analytical approach suggests a significantly smaller absolute hedge size (€356M vs €459M). This is perfectly consistent with Task 3: since the proxy mathematical formula underestimates the true portfolio loss, it naturally assumes we need a smaller IRS notional to hedge it.

Task 5: Coarse-Grained Bucket DV01 (10y, 15y)

In this task, we evaluate the portfolio's sensitivity to specific regions of the yield curve using macro-buckets. Following the provided scheme, we apply linearly interpolated weights to shift the market instruments (+1 bp), re-bootstrap the curve, and reprice the portfolio.

For the 10-year bucket, the shock weight is 1.0 up to 10 years and drops linearly to 0.0 at 15 years. For the 15-year bucket, the weight is 0.0 up to 10 years, rises linearly to 1.0 at 15 years, and remains constant at 1.0 for all longer tenors.

Description	**Result**
10y Bucket DV01	€ -923,432.52
15y Bucket DV01	€ 555,940.03

Commentary:
The 10y bucket DV01 is very negative because it mainly affects the 10-year IRS, which loses money when rates go up.

On the other hand, the 15y bucket DV01 is highly positive. The 10-year IRS is not affected by this shock because it expires before 15 years. Instead, this 15y bucket perfectly hits the swaption's underlying swap (which runs from year 10 to 15), creating a large profit.

Task 6: Delta-Hedge with Two Liquid IRS

In this task, we hedge the portfolio using two liquid instruments: a 10-year IRS and a 15-year IRS. We chose exactly these two swaps because their maturities perfectly match the 10y and 15y coarse-grained buckets defined in the previous task. To find the correct notionals, we solve a 2×2 linear system to neutralize both bucket DV01s simultaneously.

Description	**Result**
10y IRS Hedge Notional	€ -1,154,000,000 (Payer)
15y IRS Hedge Notional	€ 517,000,000 (Receiver)
Residual 10y Bucket DV01	€ 336.00
Residual 15y Bucket DV01	€ -249.00

Commentary: To protect the portfolio against non-parallel curve shifts, we neutralize the risk in both macro-buckets. We enter into a 10-year payer IRS and a 15-year receiver IRS to neutralize the sensitivities. This strategy is highly effective, leaving only tiny residual DV01s caused by the €1M rounding constraint.

Task 7: P&L Under Curve Flattening

In this final task, we test both hedging strategies under a curve flattening scenario (+1bp for 10y, -1bp for 15y). We calculate the Profit and Loss (P&L) for the portfolio hedged with only the 10-year IRS (Strategy A) and the portfolio hedged with both the 10-year and 15-year IRS (Strategy B).

Description	Result (P&L)
Strategy (A) - Single 10y IRS Hedge	€ -1,111,974.88
Strategy (B) - Two IRS Hedge	€ 1,202.53

Commentary:
Best Strategy: Strategy (B) is clearly superior. It successfully protects the portfolio against non-parallel curve movements, like this flattening scenario. Strategy (A) suffers a massive loss because it only neutralizes parallel shifts, leaving the portfolio exposed to curve reshaping.

More Accurate Strategies: We can improve accuracy by using more granular buckets (for example: 2y, 5y, 10y, 20y, 30y). This approach requires adding one liquid hedging instrument for each new bucket and solving a larger linear system.

Pros and Cons: The main advantage is higher precision. It provides better protection against complex curve movements that a simple two-bucket model might miss. The main disadvantage is the cost. Trading many different IRS contracts increases transaction costs (such as bid-ask spreads) and makes the portfolio more complex to manage and rebalance.

Identified Errors

In accordance with the assignment instructions, we identified and corrected several errors in the provided template files (`ex1_notebook` and `ex1_utilities`).

Errors in `ex1_notebook`

- **Data Initialization:** A few mistakes were identified. First, the months component of the swaption expiry was incorrectly set to zero, whereas it should be `swaption_maturity_m = 1`. Second, we have a receiver IRS, so the correct assignment is `irs_type = SwapType.RECEIVER` rather than `SwapType.PAYER`.
- **Question 2 (DV01 Calculation):** After shifting the rates of all market instruments upward by one basis point and performing a new bootstrapping, the updated discount factors (`discount_factors_up`) and the new forward swap rate (`fwd_swap_rate_up`) are correctly found. However, the value `swaption_price_up` was incorrectly computed using the previous `fwd_swap_rate` instead of the new `fwd_swap_rate_up`. Likewise, when computing `irs_mtm_up` through the function `swap_mtm`, the updated discount factors `discount_factors_up` must be used instead of the original ones.
- **Question 4 (Hedging Consistency):** The following expression contained an omission:

  ```
  delta_hedge_dv01 = (
      swaption_notional * swaption_price_up
      + delta_hedge_swap_notional * irs_mtm_up
  ) - ptf_mtm
  ```

  The new portfolio consists of the original payer swaptions, the original receiver IRSs, and an additional amount of receiver IRS used for hedging. Therefore, the term `irs_notional` × `irs_mtm_up` must also be included to maintain consistency.
- **Pricing Date:** When computing the prices, the implementation incorrectly used `today` instead of the `settlement_date`.

Errors in `ex1_utilities`

- `swaption_price_calculator`: Two issues appear in the following block:

```
if swaption_type == SwapType.PAYER:
    price = bpv * (strike * norm.cdf(-d2) - S0 * norm.cdf(-d1))
    delta = bpv * (norm.cdf(d1) - 1.0)
elif swaption_type == SwapType.RECEIVER:
    price = bpv * (S0 * norm.cdf(d1) - strike * norm.cdf(d2))
    delta = bpv * norm.cdf(d1)
```

  The swaption types were inverted (payer and receiver cases were swapped). Additionally, the formulas omitted the multiplication by the discount factor $B(t_0, t_\alpha)$, which must be included in both the price and the delta.

- `swap_mtm`: The following block contained reversed signs:

```
if swap_type == SwapType.RECEIVER:
    multiplier = 1
elif swap_type == SwapType.PAYER:
    multiplier = -1
```

  Since the function returns `multiplier` $\cdot$ (`float_leg` $-$ `fixed_leg`), the correct convention is the opposite: in a payer swap, the fixed leg is paid (multiplier $+1$), while in a receiver swap, the fixed leg is received (multiplier -1). For simplicity, our version does not use the variable multiplier.